UNIT – I: Introduction

AI problems, the foundations of AI and history of AI, Intelligent agents: Agents and Environments, the concept of rationality, the nature of environments, the structure of agents, problem solving agents, problem formulation.

Artificial "Artificial" refers to something created or produced by human effort, not occurring naturally.

- **Man-Made:** Designed or constructed by humans.
- **Synthetic:** Made through human processes, not naturally occurring.
- **Simulated:** Imitates natural phenomena via human effort.

Examples:

- **Artificial Light:** Human-created light sources.
- **Artificial Sweeteners:** Synthetic sugar substitutes.
- **Artificial Turf:** Man-made surface resembling grass.

Intelligence The ability to acquire, understand, and apply knowledge and skills.

Types of Intelligence:

- **Cognitive Abilities:** Learning, problem-solving, reasoning.
- **Emotional Intelligence:** Understanding and managing emotions.
- **Practical Intelligence:** Applying knowledge in real-world contexts.
- **Social Intelligence:** Effectively managing social interactions and relationships.

Artificial Intelligence (AI)

Definition:
AI combines "artificial" (man-made) and "intelligence" (thinking power). It refers to machines that simulate human cognitive functions such as learning, decision-making, and reasoning.

John McCarthy's Definition (Father of AI):
"The science and engineering of making intelligent machines, especially intelligent computer programs."

Other Definitions:

- "A branch of computer science dealing with the simulation of intelligent behavior in computers."
- "AI emphasizes the creation of intelligent machines that work and react like humans."

AI Problems:

1. **Bias and Fairness**
    - AI systems can inherit biases from training data, leading to unfair outcomes.
    - **Examples:** Facial recognition misidentifying certain races, biased hiring algorithms.

2. **Transparency and Explainability**
    - Many AI models, especially deep learning ones, are "black boxes," making their decision-making processes unclear.
    - **Examples:** Unclear medical diagnoses by AI, opaque loan approval decisions.
3. **Data Privacy and Security**
    - AI often requires large amounts of personal data, raising privacy and security concerns.
    - **Examples:** Personal data misuse, surveillance system ethical issues.
4. **Ethical Concerns**
    - The deployment of AI raises ethical questions about its societal impact.
    - **Examples:** Autonomous weapons, AI in law enforcement, deepface manipulation.
5. **Job Displacement**
    - AI automation may lead to job losses in certain industries.
    - **Examples:** Automation in manufacturing, customer service, and transportation.
6. **Robustness and Safety**
    - Ensuring AI systems are reliable and safe, particularly in critical applications.
    - **Examples:** Autonomous vehicles handling unexpected situations.
7. **Scalability and Efficiency**
    - Developing AI models that are computationally and energy-efficient.
    - **Examples:** High computational and energy costs in training large AI models.
8. **Generalization and Transfer Learning**
    - Ensuring AI systems can apply knowledge to new, unseen data, and transfer learning between domains.
    - **Examples:** AI models underperforming on data outside their training set, challenges in transfer learning.

The **foundations of AI** are built on several disciplines:

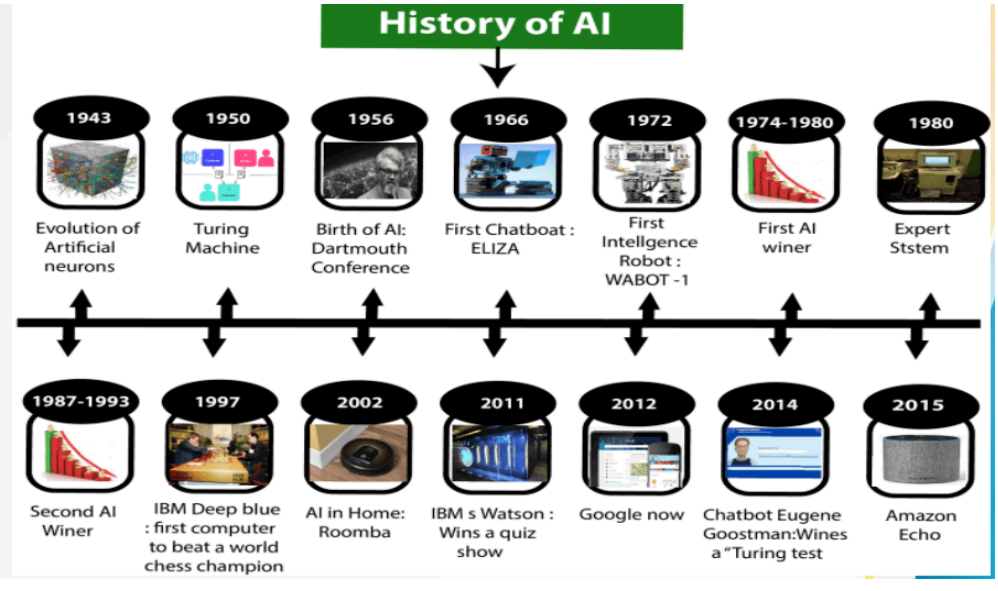
1. **Philosophy:** Provides logic and reasoning principles, addressing ethics and consciousness.
2. **Mathematics:** Offers formal logic, algorithms, and computational theory.
3. **Economics:** Influences decision-making models and rational agents.
4. **Neuroscience:** Inspires neural networks by studying brain functions.
5. **Psychology:** Provides insights into cognition and human-like learning.
6. **Linguistics:** Helps with language understanding and natural language processing.
7. **Computer Science:** The core of AI, focusing on algorithms and programming.
8. **Control Theory & Cybernetics:** Guides system behavior and feedback mechanisms.
9. **Statistics:** Enables data analysis, pattern recognition, and probabilistic reasoning.

History of AI: A Journey of Development

1. **Maturation of AI (1943-1952):**
    - **1943:** Warren McCulloch and Walter Pitts proposed the first model of artificial neurons, where each neuron was either “ON” or “OFF.”

- **1949:** Donald Hebb introduced Hebbian learning, a rule for modifying connection strengths between neurons.
- **1950:** Alan Turing published *"Computing Machinery and Intelligence"*, proposing the Turing Test to assess a machine's ability to exhibit human-like intelligence.

2. **The Birth of AI (1952-1956):**
   - **1955:** Allen Newell and Herbert A. Simon developed *"Logic Theorist"*, the first AI program, which proved 38 of 52 mathematical theorems.
   - **1956:** John McCarthy coined the term "Artificial Intelligence" at the Dartmouth Conference, marking AI as an academic field.
3. **The Golden Years - Early Enthusiasm (1956-1974):**
   - **1966:** Joseph Weizenbaum created *ELIZA*, the first chatbot.
   - **1972:** Japan developed *WABOT-1*, the first intelligent humanoid robot.
4. **The First AI Winter (1974-1980):**
   - Due to a funding shortage and diminished public interest, AI research faced stagnation, a period known as the "AI Winter."
5. **A Boom in AI (1980-1987):**
   - **1980:** AI saw a resurgence with the introduction of *Expert Systems* that emulated human expert decision-making.
   - **1980:** The first national conference of the American Association of AI was held at Stanford University.
6. **The Second AI Winter (1987-1993):**
   - Due to the high cost and limited efficiency of expert systems, funding for AI research again decreased, leading to the second "AI Winter."
7. **The Emergence of Intelligent Agents (1993-2011):**
   - **1997:** IBM's Deep Blue defeated world chess champion Gary Kasparov.
   - **2002:** Roomba, the first AI-enabled vacuum cleaner, entered homes.
   - **2006:** AI entered the business world with companies like Facebook, Twitter, and Netflix adopting AI technologies.
8. **Deep Learning, Big Data, and AI's Rise (2011-present):**
   - **2011:** IBM's Watson won *Jeopardy!*, proving its ability to understand natural language and solve complex questions.
   - **2012:** Google launched *Google Now*, a predictive Android app.
   - **2014:** The chatbot *Eugene Goostman* won the Turing Test competition.
   - **2018:** IBM's *Project Debater* debated complex topics with human experts, and Google introduced *Duplex*, a virtual assistant that made a hairdresser appointment by phone, fooling the human on the other end.



Concept of rationality / different approaches to AI

Definition:
Rationality is the ability to make decisions or take actions that are logically consistent and aimed at achieving specific goals optimally.

Key Points:

- **Rational Agent:** Acts in a way that maximizes chances of success based on available information.
- **Considerations:** Current situations and potential future outcomes.

Approaches to AI

1. Thinking Humanly

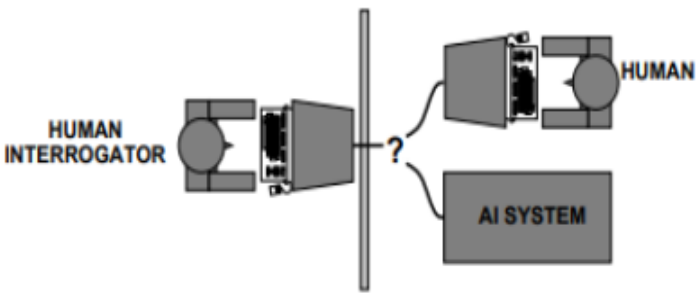
- **Focus:** Create systems that simulate human thought processes.
- **Goal:** Understand and mimic human language, emotions, and culture for natural interaction.
- **Applications:** Conversational AI, emotion recognition.
- **Examples:**
    - **Siri, Alexa, Google Assistant:** Virtual assistants using natural language processing (NLP).
    - **Emotion Recognition Systems:** Detect emotions in speech or facial expressions.

2. Acting Humanly

- **Focus:** Develop systems that act like humans.
- **Goal:** Perform tasks (e.g., speech or image recognition) in a human-like manner.
- **Applications:** Computer vision, robotics.
- **Examples:**
    - **Self-Driving Cars:** Navigate roads like humans.
    - **Facial Recognition Systems:** Identify people using facial features.
    - **Gesture-Based Interfaces:** Interpret and respond to gestures.

Turing Test (Alan Turing):

- Used to evaluate an AI's ability to act like a human.
- If a human evaluator cannot distinguish between responses from a machine and a human, the AI passes the test.



3. Thinking Rationally

- **Focus:** Build systems that reason logically.
- **Goal:** Solve problems and make decisions based on logical reasoning.
- **Applications:** Decision-making, planning, problem-solving.
- **Examples:**
    - **Expert Systems:** Provide advice based on rules and knowledge.
    - **Recommendation Systems:** Offer personalized suggestions based on user behavior.
    - **Optimization Algorithms:** Find optimal solutions considering various factors.

4. Acting Rationally

- **Focus:** Develop systems that act rationally.
- **Goal:** Make effective decisions to achieve objectives efficiently.
- **Applications:** Autonomous agents, decision-making systems.
- **Examples:**
    - **Autonomous Agents:** Make efficient decisions to achieve goals.
    - **Reinforcement Learning Algorithms:** Learn through rewards and punishments.
    - **Game AI:** Play strategic games like chess, Go, or poker.

Why Artificial Intelligence?

- **Problem-Solving:** AI solves real-world problems accurately and efficiently.
- **Virtual Assistants:** Power tools like Siri, Google Assistant, and Cortana for personalized interactions.
- **Robotics:** Enables robots to work in hazardous environments like disaster zones or deep-sea exploration.

Goals of AI

1. **Replicate Human Intelligence:** Mimic reasoning, learning, and decision-making.
2. **Solve Complex Tasks:** Address knowledge-intensive problems (e.g., financial analysis, legal research).
3. **Integrate Perception & Action:** Connect recognition (e.g., images) to actions (e.g., navigation).
4. **Perform Intelligent Tasks:** Prove theorems, play games, assist in surgery, or drive autonomously.
5. **Self-Learning Systems:** Create adaptive systems that learn, demonstrate reasoning, and advise users.

Applications of Artificial Intelligence (AI):

1. **Natural Language Processing (NLP):**
    - AI systems that understand and generate human language.
    - **Examples:**

        - Virtual assistants (Siri, Alexa)
        - Sentiment analysis of customer reviews
        - Machine translation
2. **Computer Vision:**
    - AI systems that analyze and interpret visual information.
    - **Examples:**
        - Facial recognition for security
        - Image classification
        - Object detection in autonomous vehicles
3. **Robotics:**
    - AI systems integrated with robots to interact with the physical world.
    - **Examples:**
        - Self-driving cars
        - Industrial robots
        - Service robots in hospitals
4. **Recommender Systems:**
    - AI that provides personalized suggestions based on user preferences.
    - **Examples:**
        - Netflix movie recommendations
        - Amazon product recommendations
        - YouTube video suggestions
5. **Fraud Detection:**
    - AI systems that identify fraudulent activities across various sectors.
    - **Examples:**
        - Credit card fraud detection
        - Anti-money laundering systems
        - Fraud detection in insurance claims
6. **Healthcare:**
    - AI that supports medical diagnosis, treatment planning, and drug discovery.
    - **Examples:**
        - Medical diagnosis systems
        - Drug discovery
        - Personalized medicine
7. **Customer Service:**
    - AI-powered chatbots and virtual assistants for customer support.
    - **Examples:**
        - Chatbots for customer inquiries
        - Virtual assistants for services
        - AI for triaging and routing customer service tasks
8. **Financial Services:**
    - AI systems used in financial decision-making, risk management, and fraud detection.
    - **Examples:**
        - Investment analysis tools
        - Credit scoring systems
        - Fraud detection in financial transactions
9. **Gaming:**

    - AI systems that play games and strategize, often outperforming human players.
    - **Examples:**
        - AI for chess, Go, and poker
        - AI for video game character control
        - AI for game strategy

10. **Predictive Maintenance:**

- AI that predicts equipment failures and schedules maintenance in industrial settings.
- **Examples:**
    - AI predicting equipment failures in factories
    - Maintenance scheduling systems
    - AI in power plants for critical infrastructure maintenance

Types of Artificial Intelligence

Type-1 AI (Based on Capabilities):

1. **Narrow AI (Weak AI):**
    - This is AI designed to perform a specific task.
    - It cannot perform tasks outside its predefined scope or programming.
    - Examples: Virtual assistants like Siri, Alexa, or chatbots.
2. **General AI:**
    - This refers to an AI system with general intelligence capable of performing any intellectual task that a human can do.
    - It can learn, understand, and apply knowledge across a wide range of tasks.
    - General AI remains largely theoretical and under research.
3. **Strong AI (Super AI):**
    - This AI surpasses human intelligence in virtually all fields, including problem-solving, creativity, and decision-making.
    - It is hypothetical and currently non-existent, representing the ultimate goal of AI development.

Type-2 AI (Based on Functionality):

1. **Reactive Machines:**
    - These AI systems can only react to present scenarios and cannot store past data or use it for future decisions.
    - Example: IBM's Deep Blue chess-playing computer.
2. **Limited Memory:**
    - These systems can store past experiences temporarily and use them to make decisions.
    - Example: Autonomous vehicles that observe road conditions and traffic.
3. **Theory of Mind:**
    - This level of AI would understand human emotions, beliefs, and thought processes.
    - It is under development and represents a significant step toward more human-like AI.
4. **Self-Awareness:**

    - The most advanced form of AI, where machines have their own consciousness and are aware of their existence.
    - This is purely theoretical and poses significant ethical considerations.

AI Agents

In artificial intelligence, **agents** are entities that perceive their environment and take actions based on those perceptions. The agent consists of two main components:

1. **Architecture**: The physical machinery or platform on which the AI agent operates.
2. **Agent Program**: The software or algorithm that runs on the architecture, implementing the agent function that dictates how the agent behaves.

Together, the **agent** is a combination of the architecture and the program that defines its decision-making process.

Key Concepts

1. Agent Function

- The **agent function** is a mapping from a **percept history** (the sequence of perceptions the agent has encountered) to an **action**.
- **Formal Representation**: f: P* → A, where:
    - P*: Percept sequence (the history of sensory inputs)
    - A: Set of possible actions
- The agent function helps determine the action an agent should take based on its perception history.

2. Agent Program

- The **agent program** implements the agent function. It runs on the physical architecture (hardware) to produce the agent function's mapping.

Rational Agents

A **rational agent** is one that selects actions based on its **percept history** and **built-in knowledge** to maximize its expected performance according to a predefined **performance measure**.

Rationality in an agent is defined by the following components:

- **Performance Measure**: How the agent's success is evaluated.
- **Environment**: The surroundings in which the agent operates.
- **Sensors**: The mechanisms through which the agent perceives the environment.
- **Actions**: The possible actions the agent can perform.
- **Percept Sequence**: The sequence of all perceptions the agent has experienced.

The agent is considered rational if, for each percept sequence, it selects the action expected to maximize its performance, given the evidence available.

Rationality vs. Omniscience and Perfection

- **Omniscience**: A rational agent doesn't need to know everything. For example, in a card game, it doesn't know the cards held by other players, but it still makes decisions based on what it can perceive.
- **Perfection**: Rationality is not about being perfect. A rational agent maximizes the expected outcome, while perfection is about achieving the best possible result. A rational agent can still make mistakes due to limited information.

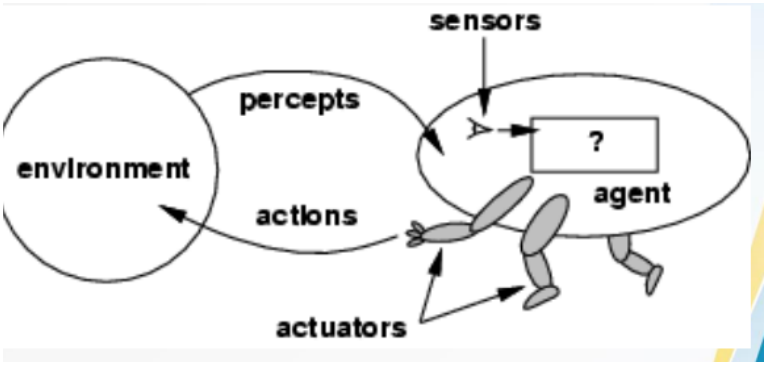
PEAS Framework

The **PEAS** framework helps define the components required to design an intelligent agent:

- **P**: **Performance Measure** – Criteria for evaluating the agent's success.
- **E**: **Environment** – The surroundings in which the agent operates.
- **A**: **Actuators** – Mechanisms the agent uses to affect the environment.
- **S**: **Sensors** – Mechanisms the agent uses to perceive the environment.

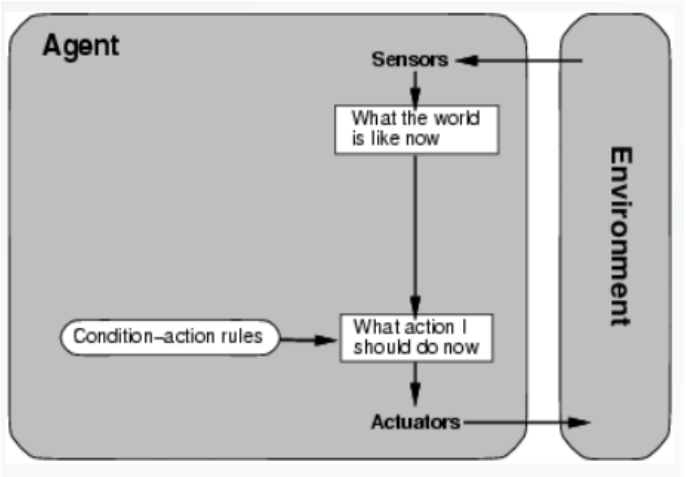
Example of PEAS for Various Agents

1. **Automated Taxi Driver**
    - **Performance Measure**: Safe, fast, legal, and comfortable trip; maximize profits.
    - **Environment**: Roads, other vehicles, pedestrians, passengers.
    - **Actuators**: Steering wheel, accelerator, brake, signals, horn.
    - **Sensors**: Cameras, sonar, GPS, speedometer, engine sensors.
2. **Part-Picking Robot**
    - **Performance Measure**: Percentage of parts placed in the correct bins.
    - **Environment**: Conveyor belt, parts, bins.
    - **Actuators**: Robot arm, grippers.
    - **Sensors**: Cameras, proximity sensors, force sensors.
3. **Interactive English Tutor**
    - **Performance Measure**: Maximize the student's score on tests.
    - **Environment**: Students using the system.
    - **Actuators**: Screen displays (exercises, suggestions, corrections).
    - **Sensors**: Keyboard (for input), speech recognition, or user interaction data.
4. **Robotic Agent (e.g., Autonomous Robot)**
    - **Performance Measure**: Complete the task (e.g., cleaning, exploration).
    - **Environment**: Workspace, objects to interact with.
    - **Actuators**: Motors for movement, manipulators.
    - **Sensors**: Cameras, infrared range finders, accelerometers.

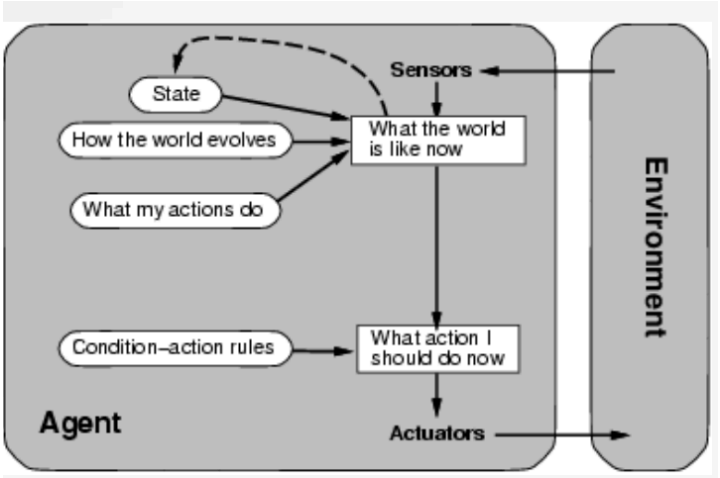


Types of AI Agents (Summary)

1. **Simple Reflex Agents**:
    - **Behavior**: React to current percepts without memory of past actions.
    - **Limitations**: No memory, leading to possible infinite loops (e.g., a vacuum cleaner that doesn't remember its location).
    - **Solution**: Randomizing actions to avoid loops.
    - **Example**: Thermostat or simple game strategies (e.g., chess openings).

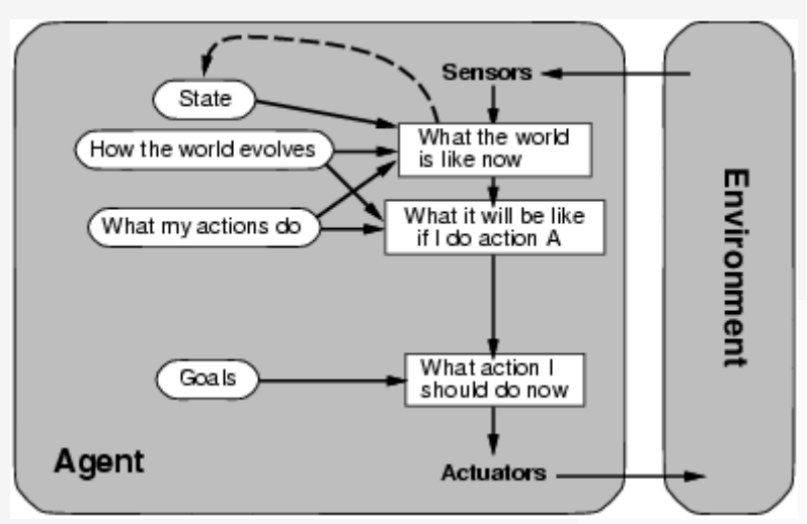


2. **Model-Based Reflex Agents**:
    - **Behavior**: Maintain an internal state to track the environment's history.
    - **Function**: These agents use a **model** of the world and update their internal state based on percept history.
    - **Example**: A car that tracks its environment to react intelligently (e.g., noticing a car is overtaking from behind).

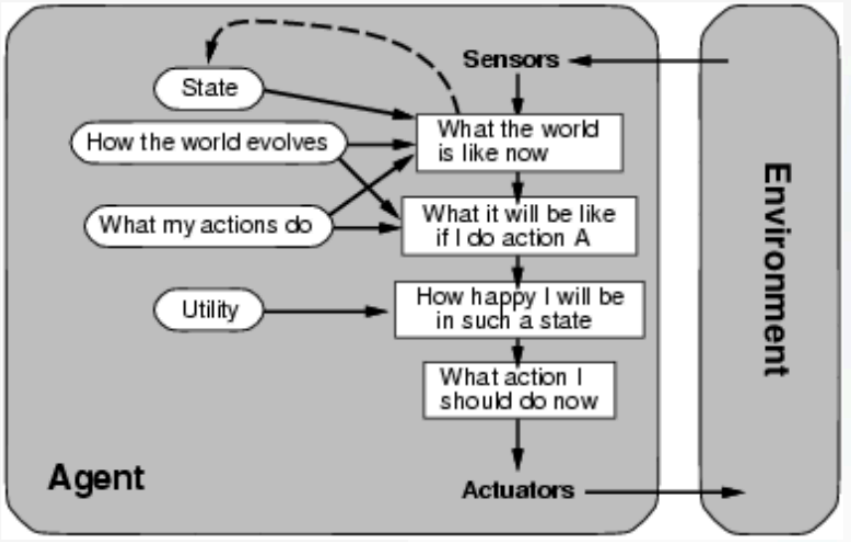


3. **Goal-Based Agents**:
    - **Behavior**: Act to achieve specific goals, using knowledge and reasoning to select actions.

- **Function**: Agents consider a sequence of actions (search and planning) to reach the goal.
- **Example**: A taxi driver deciding on the best route to a destination.

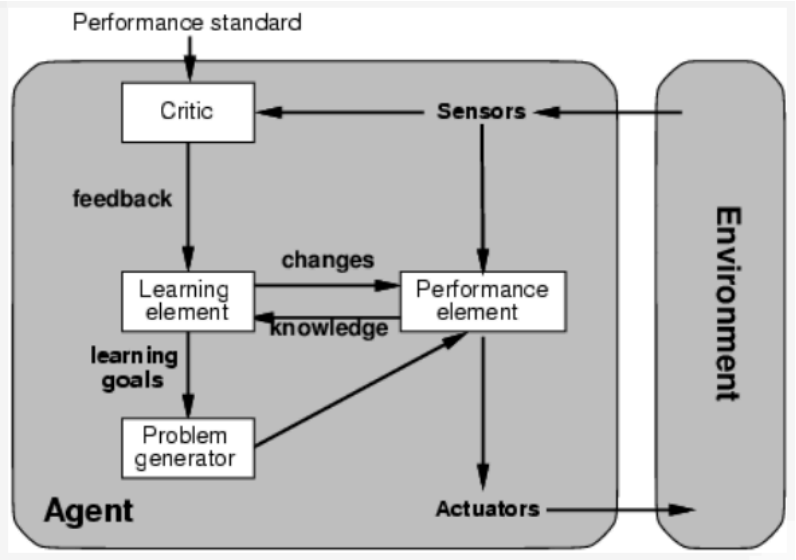


4. **Utility-Based Agents**:
   - **Behavior**: Not only aim to achieve goals but also optimize for multiple factors, such as speed, safety, and comfort.
   - **Function**: Use a utility function to evaluate and maximize happiness or success.
   - **Example**: A taxi that considers factors like speed and safety, not just reaching the destination.



5. **Learning Agents**:
   - **Behavior**: Adapt and improve based on experience.
   - **Components**:

        - **Performance Element**: Executes actions based on current knowledge.
        - **Learning Element**: Updates the performance element.
        - **Critic**: Assesses how well the agent is doing.
        - **Problem Generator**: Suggests new actions for experimentation.
    - **Example**: A taxi driver improving driving skills over time based on feedback.



Overview of Agents in AI

1. **Agent**:
    - An agent is an entity that perceives its environment and acts upon it to achieve specific goals. It makes decisions based on its perceptions and performs actions to influence the environment.
2. **Intelligent Agents**:
    - **Definition**: Autonomous entities that use sensors to perceive the environment and actuators to take actions in order to achieve their goals.
    - **Rules**:
        1. Perceives the environment.
        2. Uses perception to make decisions.
        3. Takes actions based on decisions.
        4. Actions are rational, aiming to maximize performance.
3. **Rational Agents**:
    - **Definition**: Agents that aim to maximize their performance measure by considering their perceptions and possible actions.
    - **Behavior**: Rational agents make decisions that are expected to achieve the best possible outcome, taking uncertainties into account.
4. **Human Agent**:
    - **Sensors**: Eyes, ears, nose, skin (for perceiving the environment).
    - **Effectors**: Hands, legs, mouth (for acting on the environment).
5. **Robotic Agent**:
    - **Sensors**: Cameras, infrared sensors, range finders (for perception).

    - **Actuators**: Motors, gears (for motion and control).
    - **Effectors**: Wheels, arms, grippers, etc. (for interacting with the environment).
6. **Software Agent**:
    - **Sensors**: Input streams like encoded bit strings or data (for perceiving the environment).
    - **Actuators/Effectors**: Output streams, algorithms, or processes that trigger changes in digital environments.

Environment:

- The environment is the context or external system in which an agent operates. It provides inputs (percepts) that the agent uses to make decisions and take actions that affect the environment.

Key Components of Agent Systems:

- **Sensors**: Devices that detect changes in the environment (e.g., cameras in robots, input streams in software agents).
- **Actuators**: Components that allow agents to perform actions (e.g., motors, gears).
- **Effectors**: Mechanisms that directly affect the environment (e.g., robot wheels, arms, or digital changes in software).

Characteristics of Agent-Based Systems:

- Operate within an environment.
- Perceive their surroundings via sensors.
- Act on their environment using actuators or effectors.
- Have goal-oriented behavior, focusing on achieving specific objectives.

Agent Environment in AI

The environment is a vital component in AI systems, as it provides the context in which an agent operates. It supplies sensory data and situations that require actions. The agent's effectiveness depends on how well it can perceive and interact with its environment. Here are some key characteristics of AI environments:

1. Fully Observable vs. Partially Observable

- **Fully Observable**: The agent can perceive the entire state of the environment at all times.
    - *Example*: A chess-playing agent, where the agent sees the full chessboard and all pieces.
- **Partially Observable**: The agent has limited knowledge of the environment, often due to sensor limitations or external constraints.
    - *Example*: A poker-playing agent, where it cannot see other players' cards.

2. Deterministic vs. Stochastic

- **Deterministic**: The environment's next state is completely determined by the current state and the agent's action.
    - *Example*: Chess, where every move leads to a predictable outcome.
- **Stochastic**: The environment includes random elements, and the agent cannot predict the next state with certainty.
    - *Example*: The stock market, influenced by unpredictable factors like news and market sentiment.

3. Episodic vs. Sequential

- **Episodic**: The agent's experience is divided into independent episodes, where each action is independent of others.
    - *Example*: Tic-tac-toe, where each move is based on the current state of the game, independent of past moves.
- **Sequential**: The agent's actions affect future states, and its decisions depend on the entire sequence of past actions.
    - *Example*: Chess, where each move depends on past moves and influences future ones.

4. Static vs. Dynamic

- **Static**: The environment does not change while the agent is deciding or acting.
    - *Example*: A crossword puzzle, where the environment (the puzzle itself) remains constant during the solving process.
- **Dynamic**: The environment can change while the agent is making decisions or acting.
    - *Example*: Driving a car, where traffic and road conditions change in real time.

5. Discrete vs. Continuous

- **Discrete**: The environment has a limited number of states and actions.
    - *Example*: Chess, where there are a finite number of possible moves and positions.
- **Continuous**: The environment allows for an infinite range of states and actions.
    - *Example*: A robotic arm in a factory, where movements (like adjusting angles) are continuous.

6. Known vs. Unknown

- **Known**: The agent has complete knowledge of the environment and the effects of its actions.
    - *Example*: Chess, where the agent (player) knows all possible moves and outcomes.
- **Unknown**: The agent lacks full knowledge and must explore or learn about the environment.
    - *Example*: A robot exploring an unknown planet, with no prior knowledge of the terrain.

7. Accessible vs. Inaccessible

- **Accessible**: The agent can obtain complete and accurate information about the environment's state.
    - *Example*: A room with temperature sensors, where the agent can gather real-time data.
- **Inaccessible**: The agent cannot gather complete or accurate information, affecting its decision-making.
    - *Example*: A satellite monitoring a natural disaster, lacking detailed information on certain areas.

8. Single-Agent vs. Multi-Agent

- **Single-Agent**: The agent operates alone, without interference from other agents.
    - *Example*: Solitaire, where the player (agent) interacts with the cards without other agents' influence.
- **Multi-Agent**: Multiple agents share the environment, and their actions may influence each other.
    - *Example*: A soccer game, where multiple players (agents) interact and affect each other's decisions

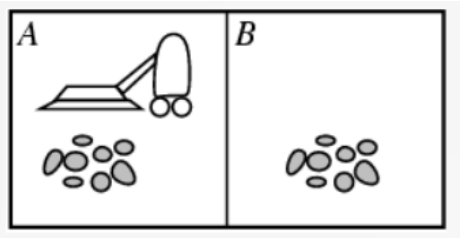
The **"Vacuum-Cleaner World"** is a classic example used in Artificial Intelligence (AI) to explain how an agent perceives its environment and acts to achieve a goal. Let's break down the PEAS (Performance Measure, Environment, Actuators, Sensors) concept for this scenario:

PEAS Framework for the Vacuum Cleaner World

The PEAS framework helps define how the vacuum cleaner operates by identifying key performance goals, understanding its environment, and specifying the mechanisms (actuators and sensors) required for efficient operation. This ensures that the AI agent performs effectively, cleaning the space while navigating obstacles and conserving energy.

1. **Performance Measure:**
    - The goal is to clean both locations (A and B) as efficiently as possible.
    - The performance is measured by:
        - The number of dirty locations cleaned.
        - The time taken (fewer steps are better).
        - Avoiding unnecessary actions.
2. **Environment:**
    - The world consists of two locations: A and B.
    - Each location can either be **clean** or **dirty**.
    - The agent knows the current location and its status (clean or dirty).
3. **Actuators (Actions):**
    - The vacuum cleaner can perform the following actions:
        - **Suck**: Clean the current location.

        - **Move Left**: Move to the adjacent location on the left.
        - **Move Right**: Move to the adjacent location on the right.
        - **NoOp**: Do nothing (used when both locations are clean).
4. **Sensors:**
    - The agent uses sensors to perceive:
        - The current location (A or B).
        - The status of the location (clean or dirty).



Agent's Function (Look-up Table):

The table below describes how the agent acts based on its percepts (current location and its status):

Explanation of Agent's Function:

- The agent observes its current location and status (clean or dirty).
- Based on the percept sequence, it takes the corresponding action from the look-up table.
- For example:
    - If the agent perceives [A, Dirty], it will clean the location by performing the **Suck** action.
    - If it perceives [A, Clean], it will move to the right

Percept Sequence	Action
[A, Clean]	Move Right
[A, Dirty]	Suck
[B, Clean]	Move Left
[B, Dirty]	Suck
[A, Clean], [A, Clean]	Move Right
[A, Clean], [A, Dirty]	Suck
[A, Dirty], [A, Clean]	Move Right
[A, Dirty], [A, Dirty]	Suck
... (similar sequences for B)	...

- **What is Artificial Intelligence?**
- **What is an agent? explain different types of agent?**
- **Explain the significance of PEAS in automated text or vaccum cleaner?**
- **Define AI and its applications?**
- **What are major History of AI?**
- **l Define the concept of rationality? OR What are four different categories of AI?**
- **l Explain A* algorithm with an example?(graph example, 8-puzzle)**
- **l Diff between greedy search and A* algorithm?**
- **l Write a short note on 1) Local beam search 2)simulated annealing search 3)DLS and uniform search cost,**
- **l Discuss about PEAS with example and explain different types also.**
- **l Discuss hill climbing features and problems?**
- **l Explain genetic algorithm with an example?**
- **l Define CSP with graph coloring.**
- **l Discuss cryptarithmetic with constraint.**
- **l Solve:**

  1) SEND + MORE= MONEY

  2) TWO + TWO= FOUR

  3) CROSS + ROADS= DANGER

  4) T0 + GO= OUT

- **l Go through few problems like TIC-TAC TOE, Water Jug Problem , 8-puzzle, graph coloring,**